INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 4: Implementation and Empirical Results

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4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one score window per freeze, one matched cohort of contracts for every same-window comparison, spender and trader cohorts for the temporal holdouts W0 and W1, and subsamples of the matched cohort for runtime against size. It reports the two single-layer walks first, their cost, their agreement in the window, their divergence, robustness checks and their holdouts (Sections 4.2 to 4.6), and then the coupled walk C-PR, the method of the thesis, under its rules and beside every other score (Section 4.7); a comparison on outcomes built from neither edge type follows (Section 4.8). It reports the revised analysis of Section 3.5.1, with amounts in US dollars, five tokens and scores from the twelve months before each freeze; this analysis was defined after the registered results were known, so no result in this chapter is a registered result. Rules A and B and R1–R3 are applied to the revised scores, with their thresholds and wording, as a re-analysis; the registered analysis reached the same verdicts except on R1, which it did not meet (Appendix 8.6, Table 8.12). Every table is generated from the evaluation summaries of Section 3.13, each Kendall τ comes with its confidence interval except in the value-slope and sample-definition checks, and each PageRank is read against the raw degree it smooths.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because one of its margin-trading protocols, GMX V2, records the position closes behind the trader labels. Each score window covers the twelve months before its freeze, 1 July 2025 to 30 June 2026 for the same-window evaluation and W1 and 1 April 2025 to 31 March 2026 for W0. The time decay is

anchored at the freeze (Figure 3.3), so the most recent six months of a window carry most of the edge weight in AWP and EndorseRank.

4.1.1 Cohorts

Four sets of contracts are used. Every address in them holds contract code; externally owned accounts (EOAs) enter the graphs as owners, senders and recipients and are never scored (Section 3.2).

- Matched cohort ($n = 6{,}483$). The contracts that received a non-zero `Approval` of one of the five tokens from at least three distinct owners in the score window of T_{obs} (Section 3.2). They are part of the analysis plan's cohort of 27,844 contracts, which applied the same rule to every token from 1 October 2023 and left out as EOAs 1,121 of its 29,022 candidates without code and 57 with an EIP-7702 delegation designator. Membership does not depend on either graph: no allowance edge touches 191 cohort contracts (by T_{obs} every allowance set in the window by them or to them was zero), no transfer edge touches $1{,}192$, and 30 are touched by neither; a contract absent from a graph scores zero under that method. Every same-window alignment, difference test and benchmark is computed on this cohort. The W0 data are drawn around a cohort of their own, the $6{,}343$ contracts that meet the same rule in the W0 score window, $5{,}383$ of which are also in the matched cohort.
- Spender cohorts of W0 and W1 ($n = 7{,}914$ and $7{,}960$). In W0 the scores are frozen at t_1 (31 March 2026, 23:59:59 UTC), and the cohort is every spender that holds at least one positive latest allowance of one of the five tokens, set inside the W0 score window, at t_1 and holds contract code: 7,914 of 7,962 such spenders. In W1 the scores are frozen at T_{obs}, and the same rule gives 7,960 contract spenders. The spender cohorts are not subsets of the matched cohort, since a spender needs only one owner with a positive allowance. The scores of these contracts are tested against owner–spender approval pairs and transfer senders first seen in the three months after each freeze (Section 3.7).
- Trader cohorts ($n = 87$ and 70). The GMX V2 accounts that hold contract code, are nodes of the freeze-date graph and close at least three positions in the label window: 87 in W0, 39 of

which had received an approval in the score window, and 70 in W1, 37 of which had. Contract accounts closed 133,685 GMX V2 positions from October 2023 to June 2026, but only 9,809 in the last six months of that span and 6,805 from July to September 2026 (Table 8.5 in Appendix 8.4.3). Every trader result therefore rests on few contracts and has wide intervals.

- Subsamples of the matched cohort ($n = 1{,}000$, $2{,}000$ and $5{,}000$). Deterministic SHA256 subsamples (seed `contract-benchmark-v1`), followed by the full cohort, used only for runtime against size and for the sensitivity to sample definition (Section 3.12.1). Each stage keeps the extracted edges incident to its sampled contracts, so its edge count grows with n.

At T_{obs} the allowance graph has 2,514,846 nodes and 3,808,949 edges, and the transfer graph 3,603,300 nodes and 8,426,124 edges. The solver graphs of Table 4.1 leave out 864 and 1,573 addresses whose pairs all carry a value weight of zero in floating point. On the same contracts the allowance graph has 45.2% as many edges as the transfer graph, because an owner–spender pair is an allowance edge only while its latest allowance of one of the five tokens, set in the window, is positive, whereas every pair of addresses that exchanged a non-zero transfer of these tokens in the window is a transfer edge. The sampling is the same for both, so the differences between the scores trace back to their edges and restarts.

The five tokens hold most of the extracted rows: of the ego-network events of the plan's cohort in the score window of T_{obs}, the revised analysis keeps 85,986,562 approvals (63.3%) and 442,722,039 transfers (76.2%), and every kept transfer has a price. Table 8.4 in Appendix 8.4.2 gives these figures per token.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two edge lists node by node (Equation (3.9)). Each evaluation run writes its scores, per-proxy statistics and bootstrap intervals to one summary file, and the benchmark its timings to another. The tables of this chapter come from runs made on 28 October 2026 from 09:16 UTC. At $d = 0.85$ every

solve converged well inside the iteration cap. Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C5)

Claim C5 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the full graphs of the matched cohort, the mean solve time and its standard deviation over five timed runs after one untimed warm-up, the peak memory of the warm-up solve, the number of iterations to convergence and the nodes and edges of the solver graph. Each graph and setting was timed once, so the same measurement reappears as the full-cohort row of Table 4.2 and as the $d = 0.85$ row of Table 4.5; Table 8.2 in Appendix 8.1.4 lists every timed run.

Table 4.1: Computational benchmark on the full graphs of the matched cohort ($n = 6{,}483$ contracts; AWP/EndorseRank runtime ratio 3.2× at an edge ratio of 2.2×). Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up, on which alone peak memory is traced; SD over the same 5 runs. A timed call turns an edge list into scores: node indexing, sparse-matrix assembly, row normalisation, restart vector and power iteration. $|V|$ and $|E|$ are the nodes and distinct edges of each method's solver graph. C-PR walks on the union of the allowance layer and the transfer layer, so its $|E|$ counts the distinct pairs of both; S-PR runs on the transfer layer, and its runtime and iterations include the solve of the allowance layer that supplies its restart vector. Table 8.2 lists every run.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$\lvert V\rvert$	$\lvert E\rvert$
EndorseRank	2.657	0.109	325.3	57	2,513,982	3,808,949
AWP	8.530	0.618	663.7	94	3,601,727	8,426,124
C-PR ($\lambda = 0.5$)	11.563	0.214	719.1	82	4,111,490	10,562,390
S-PR	10.883	0.330	509.4	151	3,603,300	8,432,635

EndorseRank solves the allowance graph in 2.66 s and AWP the transfer graph in 8.53 s, so AWP takes 3.21 times as long, at an edge ratio of 2.21. Each power iteration costs a sparse product over the $|E|$ edges and several vector operations over the $|V|$ nodes (Appendix 8.3.3). Divided by the iterations, assembly included, AWP's solve takes 0.091 s and EndorseRank's 0.047 s, 1.95 times as long, between the node ratio of 1.43 and the edge ratio, and AWP needs 1.65 times as many

iterations (94 against 57); together the two factors give the time ratio. On the same graphs that ratio is 3.00, 3.21 and 2.56 at $d = 0.75$, 0.85 and 0.95 (Section 4.5.1), so it does not follow the edge ratio. Both scores use the same solver, so the difference in speed comes from the size of each graph and from the iterations its walk needs there, which the benchmark does not separate; it is no advantage of the EndorseRank algorithm.

C-PR walks on the union of both edge sets (10,562,390 distinct pairs, 4,111,490 nodes), but its solver keeps one matrix per layer and multiplies both in every iteration, up to about 12.2 million stored entries, 1.45 times AWP's edges. It converges in fewer iterations (82 against 94) and takes 11.56 s, 1.36 times as long as AWP. S-PR runs on the transfer layer and takes 10.88 s. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector: the 151 iterations are 56 for that solve plus 95 for the seeded transfer walk. Every algorithm that walks the transfer layer, alone or with allowance edges, pays for its size.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one.

4.2.1 Runtime on subsamples of the matched cohort

Table 4.2 reports the runtimes of EndorseRank and AWP at each stage of Section 4.1.1, their ratio, the edge ratio and the nodes and edges of both solver graphs.

The stages of Section 3.12.1 change the edge ratio as well as the size: the transfer graph has 2.85 times as many edges as the allowance graph at $n = 1{,}000$, 2.95 times at $n = 2{,}000$, 2.73 at $n = 5{,}000$ and 2.21 on the full cohort. If the time of a solve followed $|E|$ alone, the time ratio would follow these edge ratios; it is 2.63, 7.49, 3.96 and 3.21. At no stage does it stay close to the edge ratio. It is 0.92 times the edge ratio at $n = 1{,}000$, where AWP needs 96 iterations and EndorseRank 43, 2.54 times at $n = 2{,}000$, where AWP needs 94 and EndorseRank 30, and 1.45 times at $n = 5{,}000$ and on the full cohort, where EndorseRank needs 57 and AWP 95 and 94; EndorseRank's solves take 0.32 to 2.66 s. The iterations each walk needs on each subgraph, and

Table 4.2: Runtime of EndorseRank (ER) and AWP against the size of the evaluation set: deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`) up to the full cohort of $n = 6{,}483$ contracts. Each stage keeps the extracted edges incident to its sampled contracts, so $|V|$ and $|E|$ of both solver graphs are reported per stage. AWP/ER time and AWP/ER $|E|$ are the ratios of the two runtimes and of the two edge counts in the same row. Runtime = mean wall-clock seconds over 5 timed solver calls after one untimed warm-up. [a]The full-cohort row is the measurement of Table 4.1, reused, not timed again.

n	ER (s)	AWP (s)	AWP/ER time	AWP/ER $\lvert E\rvert$	ER $\lvert V\rvert$	ER $\lvert E\rvert$	AWP $\lvert V\rvert$	AWP $\lvert E\rvert$
1,000	0.498	1.307	2.6×	2.8×	451,248	485,077	953,795	1,382,177
2,000	0.318	2.383	7.5×	2.9×	760,410	897,812	1,459,514	2,647,854
5,000	1.672	6.620	4.0×	2.7×	1,834,957	2,577,926	3,183,096	7,035,600
6,483[a]	2.657	8.530	3.2×	2.2×	2,513,982	3,808,949	3,601,727	8,426,124

not the edges alone, set the ratio. The stages are subsamples of one cohort graph, so they show how runtime grows with the edges up to the full graph and not beyond it. Figure 4.1 plots both runtimes against the stage size, with the edge count of each graph beside its point.

4.3 Same-window alignment (Claim C4)

Claim C4 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap confidence interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. Apart from the secondary part of rule A, fixed in the analysis plan (Section 3.7) and applied here to the revised scores as a re-analysis, none of the comparisons below was fixed in advance (Chapter 3).

The scores divide by the edges they walk on. The two that walk on transfers alone, AWP and C-PR ($\lambda = 0$), reach 0.736 ($[0.728, 0.744]$) and 0.700 ($[0.691, 0.709]$) on the transfer family and 0.226 ($[0.210, 0.245]$) and 0.194 ($[0.178, 0.213]$) on the allowance family; S-PR, whose restarts come from the allowance walk, reaches 0.671 ($[0.661, 0.681]$) and 0.289 ($[0.272, 0.308]$). The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.560 ($[0.548, 0.571]$) and 0.561

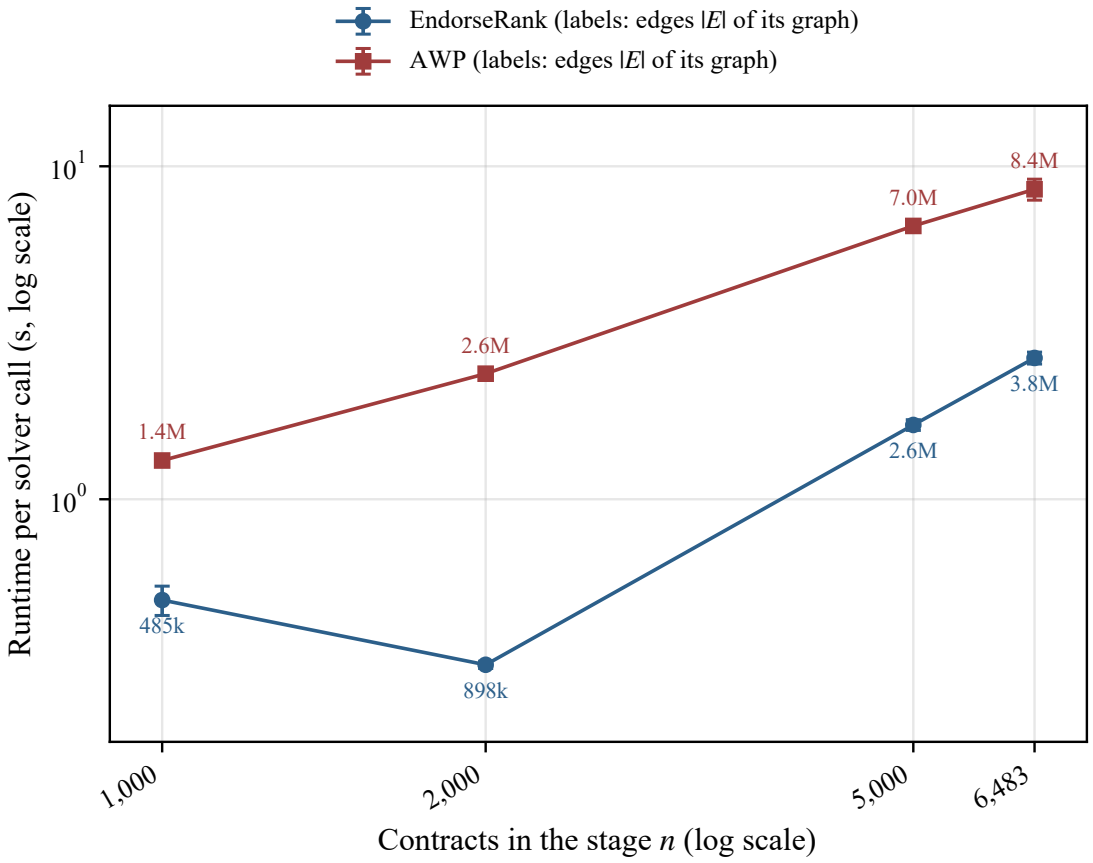


Figure 4.1: Wall-clock time of one solver call of EndorseRank and AWP against the number of contracts in each stage of Table 4.2, both axes on a log scale; the label beside each point gives the edge count $|E|$ of that graph, and error bars are the standard deviations over five timed runs.

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this study (matched cohort, $n = 6{,}483$ contracts). The scores are EndorseRank and AWP, the coupled operator C-PR at five values of λ and the endorsement-seeded walk S-PR; $\lambda = 1$ is the allowance layer walked alone (latest allowances in USD, each capped) and $\lambda = 0$ the transfer layer walked alone (USD amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families; EndorseRank with activity restarts is in Table 4.8. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.196 [0.180, 0.216]	0.560 [0.548, 0.571]	0.026 [0.012, 0.041]
AWP	0.736 [0.728, 0.744]	0.226 [0.210, 0.245]	0.419 [0.408, 0.434]
C-PR ($\lambda = 1$)	0.196 [0.180, 0.216]	0.561 [0.550, 0.572]	0.028 [0.014, 0.042]
C-PR ($\lambda = 0$)	0.700 [0.691, 0.709]	0.194 [0.178, 0.213]	0.402 [0.389, 0.417]
C-PR ($\lambda = 0.25$)	0.422 [0.407, 0.438]	0.430 [0.418, 0.444]	0.157 [0.143, 0.170]
C-PR ($\lambda = 0.5$)	0.400 [0.385, 0.416]	0.449 [0.437, 0.463]	0.142 [0.128, 0.156]
C-PR ($\lambda = 0.75$)	0.377 [0.361, 0.393]	0.464 [0.452, 0.478]	0.129 [0.115, 0.143]
S-PR	0.671 [0.661, 0.681]	0.289 [0.272, 0.308]	0.398 [0.386, 0.414]

($[0.550, 0.572]$) on the allowance family and both 0.196 ($[0.180, 0.216]$) on the transfer family, and the interior mixtures of C-PR lie between the two layers. On the Sybil-stability family the scores that walk on transfers lead, AWP with 0.419 ($[0.408, 0.434]$), while EndorseRank stays at 0.026 ($[0.012, 0.041]$). Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and in-value from the transfers and in-approve degree and value from the latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score's own edge type is therefore an intended-construct check and cannot count as an external test. The holdout of Section 4.6 asks the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP's largest coefficient is with transfer in-degree ($\tau = 0.742$, $[0.732, 0.752]$); with in-value it is almost as high (0.730, $[0.720, 0.740]$). That is what its weights lead one to expect: with the slope set at the median transfer (Section 3.6.1), an edge weight grows with the amounts transferred as well as with their number. In the registered analysis, with $b = 1$ on raw base units, V counted nearly

every transfer about once, and AWP followed the in-degree much more closely than the in-value (Appendix 8.6). It does not collapse to the in-degree either: propagation, time weights and restarts at active senders shift a measurable share of pairwise orderings, although out of window these shifts rank new senders below the in-degree (Section 4.6). On the Sybil-stability family AWP follows tenure (0.588, $[0.575, 0.600]$) and active months (0.616, $[0.603, 0.629]$), and it is positive on the inbound-counterparty ratio (0.055, $[0.033, 0.079]$).

EndorseRank follows the in-approve degree (0.650, $[0.639, 0.661]$) more closely than the in-approve value (0.469, $[0.454, 0.484]$). Its value weight is 0.8 at twice the median allowance of 100 dollars, so an allowance of a few hundred dollars and an unlimited one weigh nearly alike, whereas the in-approve value counts each allowance up to the cap of 19,000 dollars. Because every cohort contract received approvals from at least three owners, EndorseRank leaves few ties, so its coefficients rest on an ordering of nearly every contract (Section 4.4). C-PR ($\lambda = 1$) walks the same allowance graph with capped amounts, also restarts uniformly, and stays within 0.002 of EndorseRank on every family.

The first half of Claim C4 is supported in a narrow sense. Every score that walks on one edge type agrees most with the proxies of that edge type, with an interval well clear of zero, and the mixtures of C-PR lie between their two layers on every family. The agreement is partly built in. Section 4.7.1 locates C-PR within the window.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same contracts in different orders. The inter-method Kendall τ_b between the two score vectors is 0.199, with interval $[0.181, 0.221]$ (last row of Table 4.4), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Of the $21{,}011{,}403$ contract pairs, 96.4% are ordered by both scores, and 39.8% of those pairs are discordant.

The marginal score distributions are in Figure 4.2, and ties play almost no part in them: EndorseR-

ank, with $5{,}641$ distinct scores, ties 0.20% of the contract pairs, and AWP, with $5{,}292$, 3.38%. Under each score the largest tie block is formed by the contracts that score zero: 197 under EndorseRank and $1{,}192$ under AWP, that is, the 191 and $1{,}192$ contracts outside each graph and, under EndorseRank, 6 whose only pairs carry amounts too small for the value weight to differ from zero. Under EndorseRank 63 further contracts share the score $c = 1.93 \times 10^{-7}$ of a node without in-edges; 62 of them are in the allowance graph only as owners, and one has received only allowances worth less than a cent, which leave its score at c to nine significant digits; every other cohort contract has received a larger allowance and scores above c. A marginal distribution cannot show how two scores order the same contracts, so Figure 4.3 plots the two rankings against each other for a random sample of $3{,}000$ contracts. Contracts that both scores rank near the top, or both near the bottom, lie close to the identity line; in between, the points spread over most of the square.

The zero blocks come from one rule of this pipeline: a contract missing from a graph scores zero (Section 3.6). The analysis plan fixed the alternative, keeping the missing contracts as isolated nodes, as a sensitivity analysis. Under uniform teleportation an isolated node scores like any node without in-edges, so the 191 contracts outside the allowance graph receive c, the lowest score that any node of the graph can have, and join the 63 contracts at c near the bottom of the order. The 6 contracts whose only pairs carry zero weight are nodes of the graph that the solver leaves out, so they keep the score zero. No pair changes direction: the pairs between the 191 and the 63 become ties, and those between the 191 and the 6 become ordered. AWP's $1{,}192$ missing contracts were handled the same way; under AWP's restarts an isolated contract that has sent nothing receives no restart mass, so it scores zero either way. Table 4.4 gives the result. No coefficient changes at the third decimal, and the agreement between the two scores stays at 0.199 ($[0.181, 0.221]$). Because the cohort was defined by approvals received, nearly every cohort contract is a spender in the allowance graph, and the rule for missing contracts decides almost nothing.

H1 is supported. The two scores order the cohort differently, and the difference does not come from ties or from the rule for missing contracts. A protocol that relies on transfer-based scores cannot assume that its ordering of contracts survives a switch to allowance-based scores or a blend with

Table 4.4: Same-window family-mean Kendall τ with contracts missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 6{,}483$ contracts). The main analysis scores a contract that is missing from a method's graph zero; keeping it as an isolated node is the sensitivity analysis fixed in the analysis plan. The rule concerns 191 contracts for EndorseRank and 1,192 for AWP (30 are missing from both graphs). Under EndorseRank's uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated contract keeps the score zero and AWP's coefficients are the same under both rules. The last row is the agreement between the two scores. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Missing contracts scored zero		Missing contracts as isolated nodes	
Family	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.196 [0.180, 0.216]	0.736 [0.728, 0.744]	0.196 [0.180, 0.216]	0.736 [0.728, 0.744]
Allowance	0.560 [0.548, 0.571]	0.226 [0.210, 0.245]	0.560 [0.548, 0.571]	0.226 [0.210, 0.245]
Sybil stability	0.026 [0.012, 0.041]	0.419 [0.408, 0.434]	0.026 [0.012, 0.041]	0.419 [0.408, 0.434]
EndorseRank vs. AWP	0.199 [0.181, 0.221]		0.199 [0.181, 0.221]	

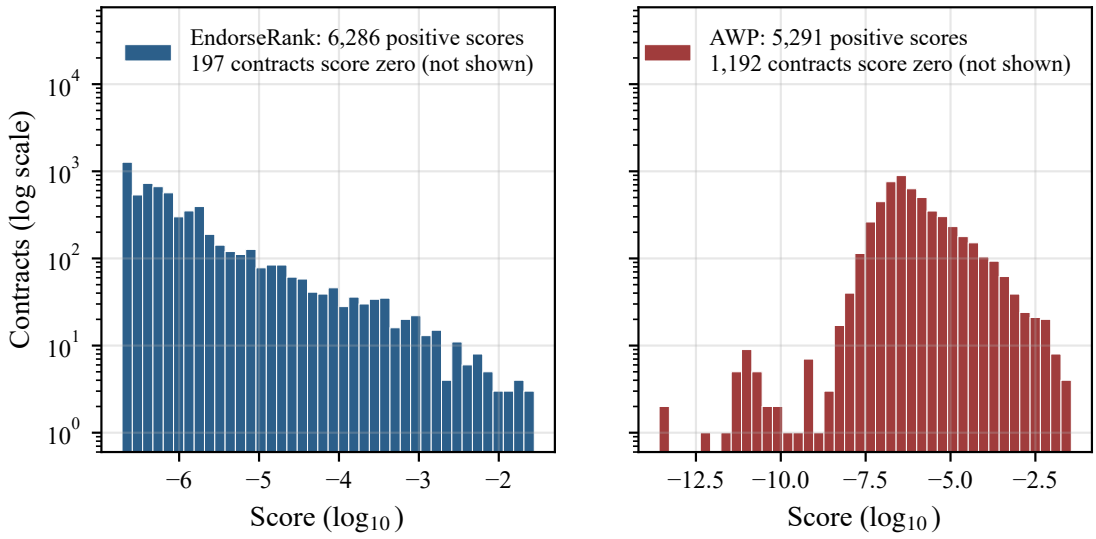


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 6{,}483$ contracts), drawn on a $\log_{10}$ scale for contracts with a positive score; counts are on a log scale. Zero-score contracts, which no edge of positive weight in the respective graph touches, are left out: 197 for EndorseRank and 1,192 for AWP.

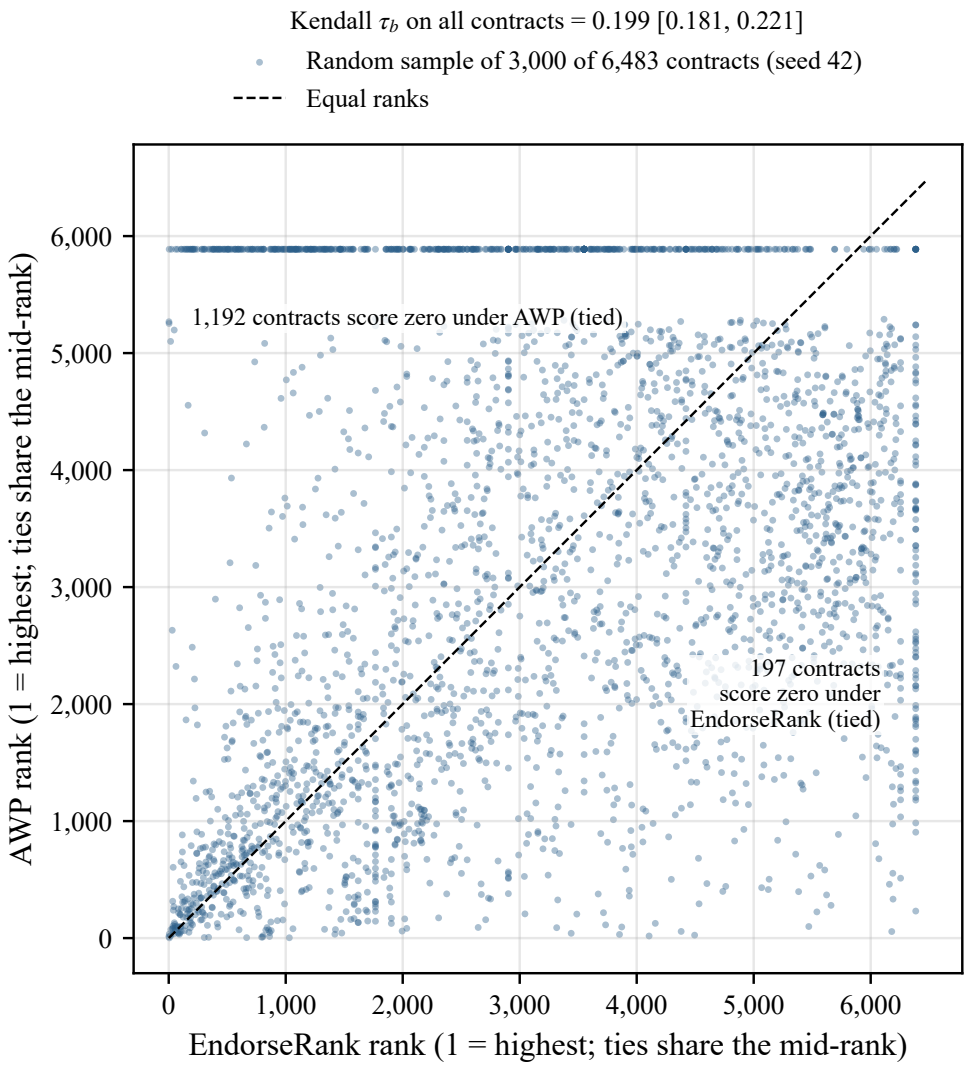


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 3,000 of the 6,483 contracts, seed 42). Tied scores share their average rank, as in Kendall's τ_b. The column at the right edge holds the sampled contracts among the 197 that score zero under EndorseRank and the row at the top those among the 1,192 that score zero under AWP. A point on the dashed identity line has the same rank in both orderings; the legend gives τ_b on all contracts with its 95% interval.

them. This divergence, the second half of Claim C4, is also the premise of the coupled walk: each single-layer walk orders the contracts by one kind of act, and Section 4.7 asks whether one walk over both edge types does better than either.

4.5 Robustness and sensitivity checks

Three checks, run for EndorseRank and AWP, ask whether the results hinge on one damping value, one slope of the value weight or one sample definition. The value-slope check takes the place of the plan's token check, which lost its point once every graph keeps five tokens (Section 3.12.3). The first two hold the sample fixed; the third varies the sample definition itself, so its coefficients are expected to move, and it is not evidence of robustness. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.5 recomputes the family means at $d \in \{0.75, 0.85, 0.95\}$ on the same graphs.

Table 4.5: Robustness: family-mean Kendall τ of EndorseRank (ER) and AWP under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched cohort ($n = 6{,}483$ contracts); all. = allowance, tr. = transfer, Syb. = Sybil stability. Runtimes are means of five timed solver calls on the full graphs; the $d = 0.85$ runtimes are the measurements of Table 4.1, reused, not timed again (Table 8.2 lists every run). CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)
0.75	0.561 [0.549, 0.573]	0.229 [0.212, 0.248]	0.196 [0.180, 0.215]	0.740 [0.732, 0.748]	0.026 [0.012, 0.041]	0.427 [0.416, 0.441]	1.986	5.956
0.85	0.560 [0.548, 0.571]	0.226 [0.210, 0.245]	0.196 [0.180, 0.216]	0.736 [0.728, 0.744]	0.026 [0.012, 0.041]	0.419 [0.408, 0.434]	2.657	8.530
0.95	0.558 [0.547, 0.570]	0.217 [0.200, 0.235]	0.197 [0.181, 0.216]	0.723 [0.715, 0.731]	0.027 [0.013, 0.041]	0.406 [0.394, 0.420]	8.113	20.805

For EndorseRank no family mean moves by more than 0.0031 across the three values. For AWP the transfer coefficient falls from 0.740 ($[0.732, 0.748]$) to 0.723 ($[0.715, 0.731]$) as d rises, and the Sybil-stability coefficient from 0.427 ($[0.416, 0.441]$) to 0.406 ($[0.394, 0.420]$). Neither score's

pattern is an artifact of the conventional $d = 0.85$. EndorseRank's stability is largely structural. For a spender whose owners receive no approvals, EndorseRank equals $c\,(1 + d\,\delta(v))$, with $\delta(v)$ the owner-normalized in-strength (Section 5.3.2), and the order that this formula gives among such spenders does not depend on d. At T_{obs} the formula holds within 1% for all 5,232 such spenders among the 6,224 cohort contracts with in-edges, so the rank correlations can hardly move. Runtime grows with the number of iterations, which rises with d: EndorseRank takes 1.99, 2.66 and 8.11 s (38, 57 and 174 iterations) and AWP 5.96, 8.53 and 20.80 s (56, 94 and 269), time ratios of 3.00, 3.21 and 2.56 at an edge ratio of 2.21. Figure 4.4 plots the family means.

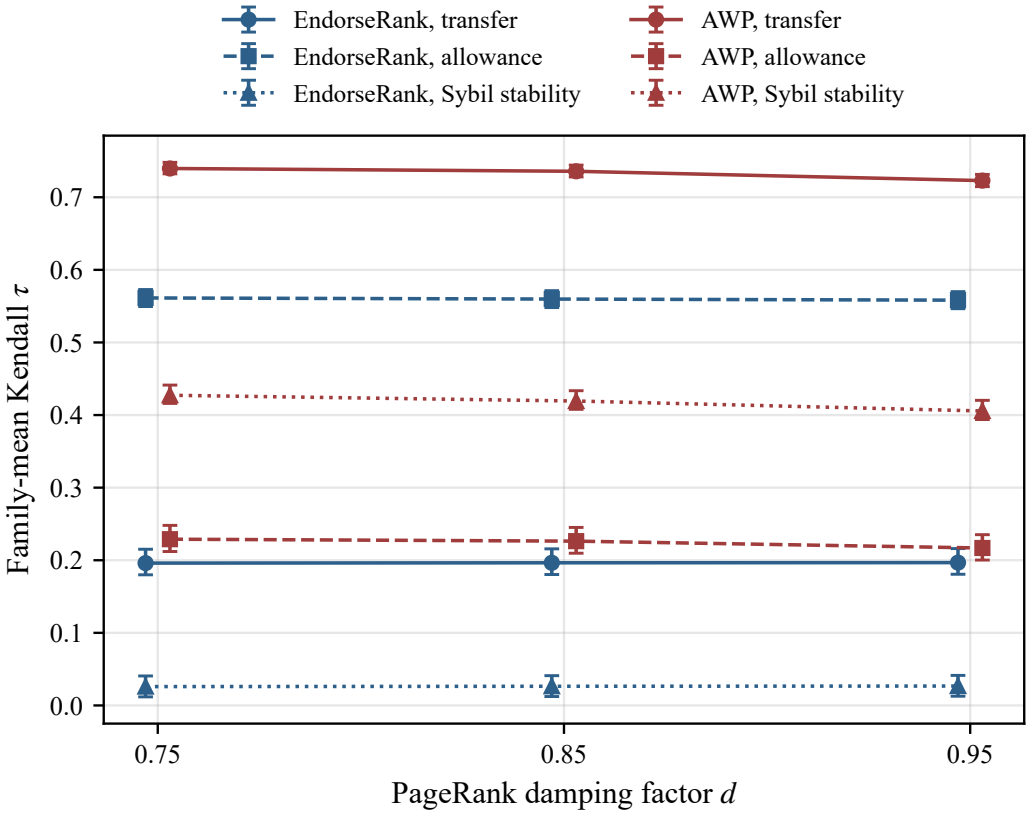


Figure 4.4: Damping sensitivity of the family-mean Kendall τ of EndorseRank and AWP on the transfer, allowance and Sybil-stability families across $d \in \{0.75, 0.85, 0.95\}$, with 95% bootstrap intervals (values from Table 4.5; matched cohort, $n = 6{,}483$ contracts).

4.5.2 Value-slope sensitivity

Table 4.6 recomputes EndorseRank and AWP at T_{obs} with the slope b of the value weight divided and multiplied by ten (Section 3.12.3), on the same graphs and proxies.

Table 4.6: Robustness: family-mean Kendall τ on the matched cohort ($n = 6{,}483$ contracts) when the slope b of the value weight V is divided or multiplied by ten, next to the main analysis. The main slopes put $V = 1/2$ at the median amount, \$170 for transfers and \$100 for allowances; $b/10$ moves that point to ten times the median and $10b$ to a tenth of it. The rank agreement τ of each variant with the main score of the same method (EndorseRank and AWP) is 0.979 and 0.969 at $b/10$, and 0.971 and 0.964 at $10b$. No intervals were computed for the variants.

Method	Transfer			Allowance			Sybil		
	$b/10$	b	$10b$	$b/10$	b	$10b$	$b/10$	b	$10b$
EndorseRank	0.196	0.196	0.196	0.561	0.560	0.555	0.026	0.026	0.027
AWP	0.736	0.736	0.734	0.224	0.226	0.226	0.421	0.419	0.419

Under either slope no family mean moves by more than 0.0046, and the pattern stays: EndorseRank leads on the allowance family, and AWP on the transfer and Sybil-stability families. The steeper slope $10b$, which puts the half weight at a tenth of the median amount and brings an edge weight closer to a count of events, lowers EndorseRank's allowance coefficient from 0.560 to 0.555 and AWP's transfer coefficient from 0.736 to 0.734; the flatter slope $b/10$ raises EndorseRank's to 0.561 and leaves AWP's at 0.736. Each variant orders the contracts almost as the main score of its method does (τ from 0.964 to 0.979). No intervals were computed for the variants.

4.5.3 Sample-definition sensitivity

Table 4.7 recomputes the family means on the subsamples of the runtime scaling.

EndorseRank moves most: its allowance coefficient is 0.665 at $n = 1{,}000$ and falls with every stage to 0.560 on the full cohort, and its transfer coefficient falls from 0.225 to 0.196. AWP stays within 0.022 of its full-cohort values on every family. No intervals were computed for the stages. None of these shifts is evidence about the scores. A stage keeps only the edges incident to its sampled contracts, so the paths that pass through unsampled cohort contracts are cut, and each

Table 4.7: Sensitivity of the family means to the evaluation set: family-mean Kendall τ of EndorseRank (ER) and AWP on deterministic SHA256 subsamples of the matched cohort (seed `contract-benchmark-v1`, full cohort $n = 6{,}483$ contracts), each stage scored on the edges incident to its contracts; all. = allowance, tr. = transfer, Syb. = Sybil stability. Rows are evaluated on different contract sets, so their τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. No intervals were computed for the stages. Runtimes are the measurements of Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER Syb.	AWP Syb.	ER (s)	AWP (s)	AWP/ER time
1,000	0.665	0.228	0.225	0.752	0.049	0.441	0.498	1.307	2.6×
2,000	0.611	0.223	0.200	0.731	0.036	0.421	0.318	2.383	7.5×
5,000	0.577	0.229	0.201	0.735	0.033	0.420	1.672	6.620	4.0×
6,483	0.560	0.226	0.196	0.736	0.026	0.419	2.657	8.530	3.2×

score comes closer to a count of the sampled contract's own edges, which is what the proxies count. We therefore report the table as a sensitivity analysis of the sample definition (Section 3.12.4), and its values must not be compared with those of Table 4.3.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). The analysis plan fixed uniform restarts for EndorseRank and named the variant as a score to report beside it, without fixing any comparison between the two. Table 4.8 scores both versions of EndorseRank, with AWP beside them.

Table 4.8: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP's rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 6{,}483$ contracts) and τ with the two spender labels in W0 ($n = 7{,}914$ spender contracts, scores frozen at 31 March 2026) and W1 ($n = 7{,}960$, frozen at 30 June 2026). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

		Same window, matched cohort			W0, spenders		W1, spenders	
Method	Restarts	Transfer	Allowance	Sybil	New approval pairs	New senders	New approval pairs	New senders
EndorseRank	uniform	0.196 [0.180, 0.216]	0.560 [0.548, 0.571]	0.026 [0.012, 0.041]	0.303 [0.287, 0.319]	0.272 [0.257, 0.286]	0.299 [0.284, 0.314]	0.259 [0.243, 0.275]
EndorseRank	by approving activity	0.251 [0.233, 0.271]	0.601 [0.590, 0.613]	0.056 [0.042, 0.072]	0.332 [0.315, 0.347]	0.270 [0.256, 0.285]	0.335 [0.319, 0.349]	0.236 [0.219, 0.251]
AWP	by sending activity	0.736 [0.728, 0.744]	0.226 [0.210, 0.245]	0.419 [0.408, 0.434]	0.196 [0.178, 0.213]	0.409 [0.396, 0.421]	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]

With AWP's restarts, EndorseRank's same-window allowance coefficient rises from 0.560 ($[0.548, 0.571]$) to 0.601 ($[0.590, 0.613]$), and its transfer coefficient from 0.196 ($[0.180, 0.216]$) to 0.251 ($[0.233, 0.271]$). The restart mass then goes to the owners that approve most, so the score weights a spender by how active its owners are as well as by how many there are. Out of window the variant ranks new approval pairs higher in both windows, and new senders no better in W0 and lower in W1. In W0 it reaches 0.332 ($[0.315, 0.347]$) on new approval pairs and 0.270 ($[0.256, 0.285]$) on new senders, $+0.029$ ($[0.023, 0.037]$) and -0.002 ($[-0.009, 0.005]$) against the uniform version, and in W1 0.335 ($[0.319, 0.349]$) and 0.236 ($[0.219, 0.251]$), $+0.035$ ($[0.029, 0.042]$) and -0.023 ($[-0.032, -0.013]$) against it. Table 4.8 gives the coefficients and Table 4.13 the W1 contrasts. These contrasts were computed after the labels were known. Neither version ranks new approval pairs above the in-approve degree: in W0 the variant falls below it (-0.060, $[-0.070, -0.052]$), as does the uniform version (-0.090, $[-0.100, -0.079]$; Section 4.6). Section 4.7.4 sets the variant beside C-PR.

4.6 Temporal holdout of the single-layer walks (RQ3)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the W0 spender cohort are frozen at t_1 (31 March 2026), and two labels are then counted for each contract between 1 April and 30 June 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.9 sets every score computed at t_1, the two raw-degree baselines included, against both labels. Table 4.10 gives the paired contrasts, first the two of rule A, fixed in the analysis plan before any log of the study was extracted, and then those computed after the labels were known. This section reads the single-layer walks and the raw degrees; rule A and the rows of C-PR and S-PR are taken up in Section 4.7.

The first finding answers the second part of RQ3 for the single-layer walks: neither walk improves on its raw degree, whatever its restart rule. AWP, which restarts at active senders, ranks new trans-

Table 4.9: Temporal holdout W0 on the spender cohort ($n = 7{,}914$ contracts): Kendall τ between scores frozen at 31 March 2026 and two labels counted on the contract between 1 April 2026 and 30 June 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the contract for the first time. The cohort keeps the 7,914 contracts among the 7,962 spenders with a positive latest allowance at the freeze; 39 EOAs without code and 9 EOAs with an EIP-7702 delegation are left out. 1,732 contracts received at least one new pair and 1,422 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.392 [0.377, 0.406]	0.284 [0.267, 0.299]
Transfer in-degree at t_1 (baseline)	0.203 [0.183, 0.224]	0.447 [0.435, 0.459]
EndorseRank	0.303 [0.287, 0.319]	0.272 [0.257, 0.286]
AWP	0.196 [0.178, 0.213]	0.409 [0.396, 0.421]
C-PR ($\lambda = 1$)	0.298 [0.282, 0.313]	0.267 [0.250, 0.282]
C-PR ($\lambda = 0$)	0.178 [0.160, 0.197]	0.366 [0.351, 0.378]
C-PR ($\lambda = 0.25$)	0.329 [0.312, 0.343]	0.345 [0.330, 0.357]
C-PR ($\lambda = 0.5$)	0.329 [0.312, 0.343]	0.336 [0.322, 0.348]
C-PR ($\lambda = 0.75$)	0.327 [0.311, 0.342]	0.327 [0.313, 0.339]
S-PR	0.190 [0.171, 0.209]	0.377 [0.363, 0.389]

Table 4.10: Paired bootstrap differences on the spender holdout W0 ($n = 7{,}914$ contracts; 400 resamples shared by a and b, seed 42). The first block holds the two contrasts of the primary part of rule A, fixed in the analysis plan committed on 9 October 2026 before the study extracted any data and applied here to the revised scores (C-PR against C-PR ($\lambda = 1$) on new approval pairs and against C-PR ($\lambda = 0$) on new transfer senders). The second block was computed after the labels were known: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1, the same comparisons for S-PR, and the increments of EndorseRank and AWP over their raw degrees; C-PR's other contrasts are in Table 4.18. Primary part of rule A (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): no worse holds, better fails, so the rule is not met. The differences between EndorseRank and AWP are in Table 8.10.

Contrast (a minus b)	Label	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Rule A, primary part, fixed in the plan						
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.329	0.298	+0.031	[0.021, 0.042]	1.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.336	0.366	-0.030	[-0.040, -0.020]	0.000
Computed after the labels were known						
C-PR ($\lambda = 1$) minus t_1 in-approve degree	new approval pairs	0.298	0.392	-0.094	[-0.104, -0.083]	0.000
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.190	0.298	-0.108	[-0.128, -0.088]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.377	0.366	+0.011	[0.004, 0.018]	1.000
EndorseRank minus t_1 in-approve degree	new approval pairs	0.303	0.392	-0.090	[-0.100, -0.079]	0.000
AWP minus t_1 in-degree	new transfer senders	0.409	0.447	-0.038	[-0.043, -0.032]	0.000

fer senders at $\tau = 0.409$ ($[0.396, 0.421]$), below the transfer in-degree at t_1 (0.447, $[0.435, 0.459]$), a paired difference of -0.038 (interval $[-0.043, -0.032]$). The transfer layer walked with uniform restarts falls further below: C-PR ($\lambda = 0$) reaches 0.366 ($[0.351, 0.378]$) (no paired contrast was computed). On the allowance side the count wins as well. In-approve degree at t_1 predicts new approval pairs with $\tau = 0.392$ ($[0.377, 0.406]$). EndorseRank reaches 0.303 ($[0.287, 0.319]$), a paired difference of -0.090 ($[-0.100, -0.079]$), and C-PR ($\lambda = 1$) 0.298 ($[0.282, 0.313]$), a difference of -0.094 ($[-0.104, -0.083]$); with activity restarts EndorseRank also stays below the count (Section 4.5.4). Within a three-month window, a spender gains more new owners and senders the more it already has, as preferential attachment would predict (Barabási & Albert, 1999), and the local count captures this directly. In EndorseRank an owner's endorsement is shared among all the spenders it approves (Section 5.3.2), so a spender whose owners approve many other spenders ranks lower than its count alone would place it; on this label that sharing lowers the coefficient. These degree contrasts were computed after the labels were known.

Read across the whole spender cohort, each label is ranked best by scores that use its own edge type. On new approval pairs the in-approve degree (0.392, $[0.377, 0.406]$) leads, followed by the C-PR mixtures (0.327, $[0.311, 0.342]$, to 0.329, $[0.312, 0.343]$), EndorseRank and C-PR ($\lambda = 1$), while the transfer in-degree (0.203, $[0.183, 0.224]$), AWP (0.196, $[0.178, 0.213]$), S-PR and C-PR ($\lambda = 0$) stay lower. On new transfer senders the transfer in-degree and AWP lead, and the allowance-side scores stay lower and close together: EndorseRank reaches 0.272 ($[0.257, 0.286]$), C-PR ($\lambda = 1$) 0.267 ($[0.250, 0.282]$) and the in-approve degree 0.284 ($[0.267, 0.299]$), with overlapping intervals. EndorseRank gives $7{,}368$ distinct scores to the $7{,}914$ spenders, and its largest tie block holds 116. Coverage does not explain the degree comparisons, but it explains part of the comparison across edge types on new approval pairs. Of the 7,914 spenders, 1,342, about one in six, have no transfer edge up to the freeze; AWP, C-PR ($\lambda = 0$) and S-PR score them zero, and only allowance edges order them. They gained new approval pairs more often than the rest (26.7% against 20.9%) and new senders almost never (1.4% against 21.3%), so AWP's zeros lower its coefficient on the first label and raise it on the second. Table 4.11 restricts the holdout to the 6,489 spenders that had received

a transfer from another address before the freeze. There AWP still ranks new senders below the transfer in-degree (0.408, $[0.393, 0.422]$, against 0.459, $[0.445, 0.474]$; -0.051, $[-0.059, -0.043]$), and EndorseRank stays below the in-approve degree on new approval pairs (0.299, $[0.283, 0.315]$, against 0.387, $[0.370, 0.402]$; -0.087, $[-0.096, -0.078]$). On new approval pairs AWP now reaches 0.328 ($[0.312, 0.343]$), above EndorseRank and below the in-approve degree (no paired contrast was computed), so EndorseRank's lead over AWP on this label rests on the 1,425 spenders left out, which include the 1,342 that AWP scores zero. The restriction was chosen after the results were known. The row of C-PR in that table is read in Section 4.7.4.

Table 4.11: Spender holdout W0 restricted to the 6,489 of the 7,914 spender contracts that had received a transfer from another address before the freeze ($n = 6{,}489$). The other 1,425 have no inbound transfer edge from another address, and 1,342 spenders score zero under AWP. The restriction was chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples; its contrasts with AWP and the in-approve degree on this subset are in Table 4.18.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.299 [0.283, 0.315]	0.315 [0.300, 0.331]
AWP	0.328 [0.312, 0.343]	0.408 [0.393, 0.422]
C-PR ($\lambda = 0.5$)[a]	0.336 [0.321, 0.351]	0.372 [0.359, 0.386]
In-approve degree at t_1	0.387 [0.370, 0.402]	0.335 [0.318, 0.350]
Transfer in-degree at t_1	0.341 [0.325, 0.357]	0.459 [0.445, 0.474]
EndorseRank minus t_1 in-approve degree (new approval pairs)	$\Delta\tau = -0.087$ [-0.096, -0.078]	
AWP minus t_1 in-degree (new transfer senders)	$\Delta\tau = -0.051$ [-0.059, -0.043]	

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these spender results bears on credit or loss; the trader labels below are built from neither edge type but are not credit labels either. Both labels are also sparse. Of the 7,914 spenders, 1,732 gained a new approval pair and 1,422 a new sender; every coefficient rests largely on how these subsets are ordered.

Other candidate labels were computed on the holdout, and they are reported here without being offered as support. Revocation counts, which for some tokens include allowances spent in full (Section 3.7), and the drain heuristic both rise with EndorseRank ($\tau = 0.343$, interval $[0.330, 0.359]$, and 0.295, interval $[0.279, 0.312]$) and with every other score, and revocations rise most steeply with

the raw in-approve degree (0.384, $[0.366, 0.399]$). For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 applies the W0 freeze to the GMX V2 contract traders and is reported as exploratory; its liquidation labels are left for the comparison of Section 4.8. Table 4.12 gives every score on the four profit labels of the 87 contract traders with at least three closes from April to June.

The walks that use transfer edges, the mixtures included, rank the number of profitable closes and the realized gain with intervals above zero: AWP reaches 0.166 ($[0.015, 0.308]$) and 0.197 ($[0.044, 0.332]$), and C-PR ($\lambda = 0$), the three mixtures and S-PR 0.163 ($[0.012, 0.289]$) to 0.172 ($[0.024, 0.300]$) and 0.273 ($[0.121, 0.397]$) to 0.297 ($[0.133, 0.427]$). The intervals of EndorseRank, its variant with activity restarts, C-PR ($\lambda = 1$) and both raw degrees contain zero, and the transfer in-degree stays low on both labels (0.069, $[-0.076, 0.210]$, and 0.074, $[-0.090, 0.229]$). The paired contrasts separate the two sides only on realized gain, for the C-PR layers (-0.257, $[-0.447, -0.071]$), and not on profitable closes (EndorseRank minus AWP -0.046, $[-0.205, 0.113]$; Table 8.11). Both labels grow with how much a contract trades. On the two shares, which do not, every coefficient is negative. On the non-loss share every interval lies below zero, from -0.167 ($[-0.297, -0.025]$) for EndorseRank to -0.288 ($[-0.450, -0.126]$) for the in-approve degree; on the profitable share none does. Part of this is built into the non-loss share (Section 4.8). None of the scores ranks how well a contract trades. The registered window gives C-PR a trader label of its own.

Table 4.12: W0 trader run: Kendall τ between scores frozen at 31 March 2026 and GMX V2 labels counted from 1 April 2026 to 30 June 2026, on the $n = 87$ GMX V2 contract accounts in the freeze-date graph with at least three closes in that window, 39 of which had received an approval by the freeze. The run is exploratory; the liquidation labels of W0 were left for the comparison (Table 4.19). Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	0.067 [-0.117, 0.225]	0.048 [-0.131, 0.222]	-0.160 [-0.338, 0.020]	-0.288 [-0.450, -0.126]
Transfer in-degree at t_1 (baseline)	0.069 [-0.076, 0.210]	0.074 [-0.090, 0.229]	-0.117 [-0.275, 0.036]	-0.214 [-0.367, -0.062]
EndorseRank	0.120 [-0.032, 0.260]	0.058 [-0.092, 0.196]	-0.047 [-0.193, 0.090]	-0.167 [-0.297, -0.025]
AWP	0.166 [0.015, 0.308]	0.197 [0.044, 0.332]	-0.121 [-0.264, 0.032]	-0.275 [-0.413, -0.145]
C-PR ($\lambda = 1$)	0.077 [-0.075, 0.233]	0.039 [-0.128, 0.197]	-0.076 [-0.240, 0.072]	-0.186 [-0.319, -0.045]
C-PR ($\lambda = 0$)	0.166 [0.020, 0.300]	0.297 [0.133, 0.427]	-0.088 [-0.247, 0.062]	-0.233 [-0.375, -0.102]
C-PR ($\lambda = 0.25$)	0.163 [0.012, 0.289]	0.289 [0.134, 0.414]	-0.082 [-0.234, 0.064]	-0.225 [-0.373, -0.089]
C-PR ($\lambda = 0.5$)	0.172 [0.024, 0.300]	0.287 [0.131, 0.413]	-0.080 [-0.227, 0.062]	-0.222 [-0.365, -0.084]
C-PR ($\lambda = 0.75$)	0.166 [0.021, 0.290]	0.273 [0.121, 0.397]	-0.076 [-0.228, 0.066]	-0.218 [-0.359, -0.081]
S-PR	0.168 [0.018, 0.301]	0.282 [0.118, 0.407]	-0.060 [-0.208, 0.092]	-0.203 [-0.354, -0.057]

4.6.2 The single-layer walks in the registered window W1

The registered window W1 (Section 3.7.2) freezes the scores at T_{obs}, 30 June 2026, and counts the same two spender labels from July to September 2026. Its registration and rule B concern C-PR and are reported in Section 4.7.3. At the freeze, 7,960 contract spenders held a positive latest allowance; 1,495 of them gained a new approval pair in the label window and 1,371 a new sender. Table 4.13 adds EndorseRank and its variant with activity restarts, which the registration does not name. None of their rows enters a rule, and the table's contrasts were computed after the labels were known, those of the variant from the saved W1 scores.

The W0 pattern holds for both single-layer walks in this window too. The in-approve degree at the freeze again ranks new approval pairs above EndorseRank (0.381, $[0.365, 0.396]$, against 0.299, $[0.284, 0.314]$; -0.082, $[-0.090, -0.073]$), and the transfer in-degree again ranks new senders above AWP (0.439, $[0.426, 0.454]$, against 0.398, $[0.386, 0.412]$; -0.041, $[-0.047, -0.034]$). With activity restarts EndorseRank still ranks new approval pairs below the degree (-0.046, $[-0.055, -0.038]$), as it did in W0. On the liquidation-free close rate of the W1 traders EndorseRank reaches 0.297 ($[0.174, 0.418]$).

4.7 The coupled walk C-PR (RQ4, RQ5)

C-PR is the method of this thesis: one random walk over the US-dollar allowance layer and the US-dollar transfer layer, which at each node mixes the two transition rows with weight λ (Section 3.6.3). The analysis plan fixed one decision rule, and its object is C-PR: rule A asks the coupled walk to beat each of its layers walked alone (RQ4). The registration of W1 added rule B, which asks it to improve on its transfer layer on new data (RQ5). Both rules compare C-PR with its own layers, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), because only that contrast changes the coupling and nothing else; C-PR also differs from EndorseRank and AWP in its amount weights and its uniform restarts. Both rules are applied to the revised scores as a re-analysis. This section places C-PR within the window, then reports rule A in W0 and rule B in W1, and ends with its contrasts with every other

Table 4.13: EndorseRank and its variant with activity restarts on the cohorts and labels of the window W1 (spenders $n = 7{,}960$ contracts, traders $n = 70$). Neither these rows nor the contrasts below them are part of rule B, and the contrasts were not named in the registration. The contrasts of EndorseRank with activity restarts were computed afterwards from the saved W1 scores. The AWP, C-PR and degree rows repeat Table 4.15, and every contrast uses the same contract resamples; the differences between EndorseRank and AWP are in Table 8.10, and those of C-PR in Table 4.18. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 7{,}960$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.381 [0.365, 0.396]	0.269 [0.250, 0.285]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.165 [0.143, 0.186]	0.439 [0.426, 0.454]	0.253 [0.116, 0.363]
EndorseRank	0.299 [0.284, 0.314]	0.259 [0.243, 0.275]	0.297 [0.174, 0.418]
EndorseRank, restarts by approving activity	0.335 [0.319, 0.349]	0.236 [0.219, 0.251]	0.286 [0.160, 0.404]
AWP	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]	0.263 [0.140, 0.379]
C-PR ($\lambda = 0.5$)	0.316 [0.300, 0.330]	0.324 [0.308, 0.337]	0.282 [0.162, 0.397]
EndorseRank minus in-approve degree, new approval pairs	$\Delta\tau = -0.082$ [-0.090, -0.073]		
AWP minus transfer in-degree, new transfer senders	$\Delta\tau = -0.041$ [-0.047, -0.034]		
EndorseRank (activity restarts) minus in-approve degree, new approval pairs	$\Delta\tau = -0.046$ [-0.055, -0.038]		
EndorseRank (activity restarts) minus EndorseRank, new approval pairs	$\Delta\tau = +0.035$ [0.029, 0.042]		
EndorseRank (activity restarts) minus EndorseRank, new transfer senders	$\Delta\tau = -0.023$ [-0.032, -0.013]		

score of the study, which decide nothing.

4.7.1 The coupled walk within the window

Two hybrids were defined in Section 3.6.3. C-PR mixes the US-dollar allowance and transfer layers in each node's transition row with weight λ, and its ablation S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.14 gives their paired contrasts in this window: first the eight of Section 3.10 against the two layers walked alone, then six against EndorseRank and AWP, two of them computed afterwards from the saved scores. RQ4 and RQ5 are decided out of window. Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

C-PR lies between its two layers on every family and stays nearer to its allowance layer. At the primary setting it reaches 0.449 ($[0.437, 0.463]$) on the allowance family, 0.112 below C-PR ($\lambda = 1$) ($[-0.124, -0.100]$), and 0.400 ($[0.385, 0.416]$) on the transfer family, 0.300 below C-PR ($\lambda = 0$) ($[-0.318, -0.281]$). As λ rises from 0.25 to 0.75 it moves toward the allowance layer on all three

Table 4.14: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the two layers they are built from, the allowance layer (C-PR ($\lambda = 1$)) and the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 6{,}483$ contracts). Family-mean τ; the same contract resamples are used for a and b. The contrasts with the layers are a stricter check beside the secondary part of rule A, which compares C-PR's point estimate with the better layer's interval in Table 4.3. The secondary part of rule A, fixed in the analysis plan and applied here to the revised scores, is not met: C-PR's point estimate lies outside the interval of C-PR ($\lambda = 0$) on the transfer family and outside the interval of C-PR ($\lambda = 1$) on the allowance family, and C-PR does not exceed both layers on the Sybil-stability family. An interval below 0 means a is below b beyond sampling variation; one above 0 means it exceeds it. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42). [a]Computed afterwards from the saved scores, with the same resamples: C-PR against EndorseRank on the transfer family and against AWP on the allowance family, the family that each single-layer walk follows least.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
Against the two layers walked alone					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.400	0.700	-0.300	[-0.318, -0.281]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	0.449	0.561	-0.112	[-0.124, -0.100]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.142	0.402	-0.260	[-0.278, -0.241]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.142	0.028	+0.115	[0.104, 0.123]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.671	0.700	-0.029	[-0.036, -0.023]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.289	0.561	-0.272	[-0.290, -0.253]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.398	0.402	-0.004	[-0.008, 0.000]	0.037
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.398	0.028	+0.371	[0.351, 0.390]	1.000
Against EndorseRank and AWP					
C-PR transfer minus AWP transfer	0.400	0.736	-0.336	[-0.354, -0.316]	0.000
C-PR allowance minus EndorseRank allowance	0.449	0.560	-0.111	[-0.122, -0.099]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.142	0.419	-0.277	[-0.294, -0.260]	0.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.142	0.026	+0.116	[0.105, 0.125]	1.000
C-PR transfer minus EndorseRank transfer[a]	0.400	0.196	+0.204	[0.191, 0.214]	1.000
C-PR allowance minus AWP allowance[a]	0.449	0.226	+0.223	[0.206, 0.240]	1.000

families. AWP ranks the transfer family better than C-PR (-0.336, $[-0.354, -0.316]$), and EndorseRank the allowance family (-0.111, $[-0.122, -0.099]$). On the family that each single-layer walk follows least, C-PR leads that walk: it ranks the transfer family 0.204 above EndorseRank ($[0.191, 0.214]$) and the allowance family 0.223 above AWP ($[0.206, 0.240]$). A node with out-edges in only one layer follows that layer whatever λ is (Equation (3.9)); λ acts only at the nodes with out-edges in both layers. Almost every node of the allowance layer is an owner with out-edges, so the allowance rows enter the walk wherever an owner also sent a transfer that the extraction kept. On the Sybil-stability family C-PR exceeds its allowance layer ($+0.115$, $[0.104, 0.123]$) and lies below its transfer layer (-0.260, $[-0.278, -0.241]$).

On every family, S-PR falls below the layer that leads that family or is level with it. Restarting the transfer walk from the allowance walk drops the transfer coefficient to 0.671 ($[0.661, 0.681]$; $\Delta\tau = -0.029$, $[-0.036, -0.023]$, against C-PR ($\lambda = 0$)), raises the allowance coefficient only as far as 0.289 ($[0.272, 0.308]$; $\Delta\tau = -0.272$, $[-0.290, -0.253]$, against C-PR ($\lambda = 1$)), and leaves Sybil-adjusted stability level with the transfer layer (-0.004, $[-0.008, 0.000]$). A restart vector changes where the walk lands after each teleport. It carries part of the allowance signal into the transfer walk, but less than walking the allowance edges does. This is the same-window counterpart of the finding that Section 4.7.2 reaches out of window.

As fixed in the analysis plan, the secondary part of rule A asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on Sybil-adjusted stability. At $\lambda = 0.5$, C-PR fails all three parts, as in the registered analysis (Appendix 8.6). Its transfer coefficient (0.400) lies below the interval of C-PR ($\lambda = 0$) ($[0.691, 0.709]$), its allowance coefficient (0.449) lies below that of C-PR ($\lambda = 1$) ($[0.550, 0.572]$), and on Sybil-adjusted stability it exceeds only the allowance layer. The paired contrasts of Table 4.14, reported alongside the rule as a stricter check, point the same way. In the same-window tables the mixture gives up part of each layer's agreement with its own family and exceeds the better layer on no family. The holdout takes up the remaining question of whether the mixture has any predictive content that the single layers lack.

4.7.2 Rule A in the window W0

The coupled walk fails rule A out of window, although in an exploratory comparison it improves on its transfer layer walked alone. That rule, applied here to the revised scores as a re-analysis, asked C-PR to do no worse than C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders in W0 (first block of Table 4.10). At $\lambda = 0.5$, C-PR lies above the allowance walk on new approval pairs ($\Delta\tau = +0.031$, interval $[0.021, 0.042]$), so the first condition holds, and below the transfer walk on new transfer senders ($\Delta\tau = -0.030$, interval $[-0.040, -0.020]$), so the second fails, and with it the primary part, as in the registered analysis (Table 8.12). Against the transfer walk alone, a comparison the rule did not include, C-PR does better. On new approval pairs C-PR reaches 0.329 ($[0.312, 0.343]$) against 0.178 ($[0.160, 0.197]$), a paired difference of $+0.151$ (interval $[0.130, 0.170]$, on the same resamples). The transfer walk, like AWP, scores zero the spenders without a transfer edge, and C-PR was not contrasted with it on the other spenders, so part of this gain may come from ordering those spenders. This contrast was computed after the labels were known; in the registered analysis the same contrast became the hypothesis F1 of the registration of W1 (Section 4.7.3). Across the three values of λ, C-PR is flat on new approval pairs (0.329, $[0.312, 0.343]$, at $\lambda = 0.25$; 0.327, $[0.311, 0.342]$, at $\lambda = 0.75$), and its point estimate on new transfer senders falls as λ rises, from 0.345 ($[0.330, 0.357]$) to 0.327 ($[0.313, 0.339]$) (no paired contrast was computed, and λ was not tuned).

The seeded ablation shows where the allowance signal has to sit. S-PR ranks the flow label slightly above the transfer walk ($\Delta\tau = +0.011$, interval $[0.004, 0.018]$) and falls below the allowance walk on the endorsement label ($\Delta\tau = -0.108$, $[-0.128, -0.088]$), ranking new approval pairs only a little above the transfer walk (0.190, $[0.171, 0.209]$, against 0.178). When transfer edges decide where the walk goes, a restart distribution derived from endorsements changes little, although, like the transfer walk, S-PR scores zero the spenders without a transfer edge, so part of this gap may come from coverage. On the spender holdout, the allowance signal survives when allowance edges carry the walk, fully in the allowance walk and mixed in C-PR, and most of it is lost when they

only decide where the walk restarts.

4.7.3 Rule B in the registered window W1

The registration of Section 3.7.2 was committed to the public repository as commit `0d3a057` at 16:49:32 UTC on 9 October 2026, after the W0 results of the registered analysis were known and before the study extracted the July–September logs. It fixed rule B with the margin $\delta = 0.01$, the half-width of the W0 interval of the F2 contrast in the registered analysis (0.0040), rounded up. The extraction manifest records no use of the testing override (Appendix 8.1). The July–September logs held 6,805 GMX V2 closes by contract accounts, all of which decoded. The revised scores were evaluated afterwards, with 400 paired resamples and seed 42, and rule B and R1–R3 are applied to them as a re-analysis; the registered analysis reached the same verdicts except on R1 (Appendix 8.6).

The spender cohort is that of Section 4.6.2. Seventy GMX V2 contract traders in the freeze-date graph closed at least three positions in the label window; 9 of them had at least one liquidation, and 1 had every close liquidated. Table 4.15 gives every score of rule B on its three labels, and Table 4.16 the contrasts with their verdicts.

The rule is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.316$ ($[0.300, 0.330]$) against 0.146 ($[0.126, 0.168]$) for its transfer layer walked alone, a difference of $+0.170$ (interval $[0.149, 0.189]$), so the W0 gain on authorization recurs. The in-approve degree at the freeze ranks this label at 0.381 ($[0.365, 0.396]$), above C-PR's point estimate, as it did in W0 (0.392 against 0.329), and C-PR lies below it in both windows (Section 4.7.4); F1 shows a gain over the transfer walk, part of which may come from the spenders that walk scores zero, not a gain over the approval count. F2 fails. On new transfer senders C-PR reaches 0.324 ($[0.308, 0.337]$) against 0.348 ($[0.335, 0.363]$); the difference is -0.025 ($[-0.038, -0.015]$), and the whole interval lies below the margin of -0.01. F3 fails narrowly. On the liquidation-free close rate of the traders C-PR reaches 0.282 ($[0.162, 0.397]$) and the transfer walk 0.282 ($[0.162, 0.394]$), and the difference is $+0.000$ ($[-0.010, 0.010]$), whose lower end lies just below -0.01. The rule needed all three

Table 4.15: Window W1: Kendall τ between scores frozen at 30 June 2026 and the three labels of rule B, counted from 1 July 2026 to 30 September 2026. The spenders are the $n = 7{,}960$ contract spenders holding a positive latest allowance at the freeze; 1,495 of them gained a new approval pair and 1,371 a new sender. The traders are the $n = 70$ GMX V2 contract accounts in the freeze-date graph with at least three closes in the label window; 9 of them had at least one liquidation and 1 had only liquidations. C-PR ($\lambda = 0$) is the comparator of rule B and AWP that of its first sensitivity analysis (Table 4.16); EndorseRank is in Table 4.13. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Spenders ($n = 7{,}960$)		Traders ($n = 70$)
Score at 30 June 2026	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.381 [0.365, 0.396]	0.269 [0.250, 0.285]	0.394 [0.261, 0.524]
Transfer in-degree at the freeze (baseline)	0.165 [0.143, 0.186]	0.439 [0.426, 0.454]	0.253 [0.116, 0.363]
AWP	0.157 [0.137, 0.178]	0.398 [0.386, 0.412]	0.263 [0.140, 0.379]
C-PR ($\lambda = 1$)	0.295 [0.280, 0.310]	0.257 [0.241, 0.274]	0.302 [0.177, 0.425]
C-PR ($\lambda = 0$)	0.146 [0.126, 0.168]	0.348 [0.335, 0.363]	0.282 [0.162, 0.394]
C-PR ($\lambda = 0.25$)	0.315 [0.299, 0.328]	0.332 [0.317, 0.346]	0.282 [0.162, 0.397]
C-PR ($\lambda = 0.5$)	0.316 [0.300, 0.330]	0.324 [0.308, 0.337]	0.282 [0.162, 0.397]
C-PR ($\lambda = 0.75$)	0.316 [0.298, 0.330]	0.315 [0.299, 0.329]	0.284 [0.160, 0.398]
S-PR	0.162 [0.140, 0.184]	0.367 [0.354, 0.381]	0.300 [0.175, 0.411]

hypotheses, so, as the registration said for this case, the thesis makes no claim that C-PR improves on a walk over the transfers alone.

The registration warned that F3 could pass with both coefficients near zero. Neither is near zero here: every score of Table 4.15 ranks this label with an interval above zero, from 0.253 ($[0.116, 0.363]$) for the transfer in-degree at the freeze to 0.394 ($[0.261, 0.524]$) for the in-approve degree, the highest point estimate; of the walks only EndorseRank and C-PR were contrasted with it (Sections 4.7.4 and 4.8).

Of the results reported beside the decision, F4 fails: on the liquidation-free close rate C-PR is not better than the transfer walk. F5 fails, since C-PR and the allowance walk cannot be told apart on that label (-0.019, $[-0.123, 0.089]$). F6, the primary part of rule A, fails again: C-PR lies above the allowance walk on new approval pairs ($+0.021$, $[0.009, 0.032]$), which meets the no-worse condition, but below the transfer walk on new senders. As in W0, C-PR's coefficients are flat across λ on new approval pairs and fall on new senders as λ rises. S-PR ranks new senders

Table 4.16: Window W1: the paired contrasts F1–F6 of rule B with their verdicts, and the two sensitivity analyses (rule B and its sensitivity analyses were registered for the raw-unit analysis; here they are applied to the revised scores). $\Delta\tau = \tau(a) - \tau(b)$, with C-PR ($\lambda = 0.5$) as a. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above $-\delta = -0.01$; F6a, the first part of rule A, holds when the interval contains zero or lies above it. F1, F2, F6a and F6b are on the spenders ($n = 7{,}960$ contracts), F3–F5 on the traders ($n = 70$), as in Table 4.15. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	Contrast	Label	Test	$\tau(a)$	$\tau(b)$	$\Delta\tau$ [95% CI]	Verdict
Rule B: C-PR ($\lambda = 0.5$) against its transfer layer walked alone, C-PR ($\lambda = 0$)							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.316	0.146	+0.170 [0.149, 0.189]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.324	0.348	-0.025 [-0.038, -0.015]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.282	0.282	+0.000 [-0.010, 0.010]	fails
Decision: F1, F2 and F3 must all hold							not met
Reported, not part of the decision							
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.282	0.282	+0.000 [-0.010, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.282	0.302	-0.019 [-0.123, 0.089]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.316	0.295	+0.021 [0.009, 0.032]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.324	0.348	-0.025 [-0.038, -0.015]	fails
Sensitivity analysis 1: AWP as the comparator							
F1	C-PR vs. AWP	new approval pairs	superiority	0.316	0.157	+0.159 [0.139, 0.178]	holds
F2	C-PR vs. AWP	new transfer senders	non-inferiority, -0.01	0.324	0.398	-0.075 [-0.085, -0.065]	fails
F3	C-PR vs. AWP	liquidation-free close rate	non-inferiority, -0.01	0.282	0.263	+0.019 [-0.026, 0.064]	fails
F4	C-PR vs. AWP	liquidation-free close rate	superiority	0.282	0.263	+0.019 [-0.026, 0.064]	fails
Sensitivity analysis 2: contracts outside a graph kept as isolated nodes							
F1	C-PR vs. C-PR ($\lambda = 0$)	new approval pairs	superiority	0.316	0.148	+0.168 [0.147, 0.187]	holds
F2	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	non-inferiority, -0.01	0.324	0.349	-0.025 [-0.039, -0.016]	fails
F3	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	non-inferiority, -0.01	0.282	0.282	+0.000 [-0.010, 0.010]	fails
F4	C-PR vs. C-PR ($\lambda = 0$)	liquidation-free close rate	superiority	0.282	0.282	+0.000 [-0.010, 0.010]	fails
F5	C-PR vs. C-PR ($\lambda = 1$)	liquidation-free close rate	superiority	0.282	0.322	-0.040 [-0.116, 0.035]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$)	new approval pairs	no worse	0.316	0.295	+0.021 [0.009, 0.032]	holds
F6b	C-PR vs. C-PR ($\lambda = 0$)	new transfer senders	superiority	0.324	0.349	-0.025 [-0.039, -0.016]	fails

at 0.367 ($[0.354, 0.381]$), above the transfer walk's point estimate, and new approval pairs at 0.162 ($[0.140, 0.184]$), below the allowance walk (0.295, $[0.280, 0.310]$); no paired contrast of S-PR was registered for W1.

Neither registered sensitivity analysis changes the verdict (Table 4.16). With AWP as the comparator, F1 still holds ($+0.159$, $[0.139, 0.178]$), a lead that comes largely, but not only, from the spenders AWP scores zero (Section 4.7.4), and F2 fails clearly: AWP ranks new senders at 0.398 ($[0.386, 0.412]$), and C-PR's difference to it is -0.075 ($[-0.085, -0.065]$), an interval that lies entirely below the margin. F3 and F4 fail ($+0.019$, $[-0.026, 0.064]$). On 70 traders this interval is 0.090 wide, so F3 could have held only with a lead of about 0.035; its failure leaves a loss of 0.026 possible and does not show that C-PR ranks the label worse. With the contracts outside a graph kept as isolated nodes, every verdict stays as it was. The point estimates change by at most 0.002, except on F5: the rule lifts the traders outside the allowance graph from zero to the score of a node without in-edges, the allowance walk's coefficient on the liquidation-free close rate rises from 0.302 to 0.322, and C-PR's difference to it widens to -0.040 ($[-0.116, 0.035]$).

Table 4.17 gives the W1 trader labels that do not enter the decision, with the 2,000 resamples of Section 4.8. On the two count labels the walks that use transfer edges again lead, while the transfer in-degree ranks both with intervals that just contain zero (0.156, $[-0.012, 0.306]$; 0.142, $[-0.018, 0.278]$), and the intervals of C-PR at $\lambda = 0.5$ and 0.75 on profitable closes, and those of the three mixtures and S-PR on realized gain, contain zero. AWP has the highest point estimates on profitable closes (0.264, $[0.103, 0.408]$) and realized gain (0.280, $[0.111, 0.435]$); its lead over C-PR ($\lambda = 0$) is resolved on neither (C-PR ($\lambda = 0$) minus AWP -0.042, $[-0.087, 0.005]$, and -0.052, $[-0.108, 0.002]$). The allowance-side scores stay near zero on both. On the no-liquidation flag every score is positive, the in-approve degree highest (0.407, $[0.268, 0.541]$). On both shares the intervals of the in-approve degree, C-PR ($\lambda = 1$), EndorseRank and its variant with activity restarts lie below zero, for the in-approve degree at -0.294 ($[-0.468, -0.106]$) on the non-loss share, while AWP and C-PR ($\lambda = 0$) are positive with intervals that contain zero.

Table 4.17: Window W1, traders: Kendall τ between scores frozen at 30 June 2026 and the other GMX V2 labels counted from 1 July 2026 to 30 September 2026, on the $n = 70$ contract accounts with at least three closes in that window. No liquidation is 1 when none of the account's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter rule B. The intervals use the resamples of Table 4.19. CI = 95% percentile bootstrap over contracts (2,000 paired resamples, seed 42).

Score at 30 June 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.407 [0.268, 0.541]	0.042 [-0.158, 0.243]	0.033 [-0.164, 0.230]	-0.221 [-0.406, -0.034]	-0.294 [-0.468, -0.106]
Transfer in-degree at the freeze (baseline)	0.266 [0.125, 0.395]	0.156 [-0.012, 0.306]	0.142 [-0.018, 0.278]	-0.059 [-0.225, 0.099]	-0.071 [-0.225, 0.082]
EndorseRank	0.303 [0.172, 0.422]	0.063 [-0.124, 0.253]	-0.011 [-0.211, 0.187]	-0.184 [-0.341, -0.014]	-0.265 [-0.411, -0.107]
EndorseRank, restarts by approving activity	0.292 [0.155, 0.415]	0.049 [-0.129, 0.237]	-0.019 [-0.195, 0.166]	-0.200 [-0.345, -0.037]	-0.275 [-0.418, -0.123]
AWP	0.265 [0.139, 0.383]	0.264 [0.103, 0.408]	0.280 [0.111, 0.435]	0.062 [-0.101, 0.213]	0.049 [-0.090, 0.185]
C-PR ($\lambda = 1$)	0.308 [0.175, 0.428]	0.009 [-0.167, 0.192]	-0.066 [-0.249, 0.120]	-0.208 [-0.370, -0.030]	-0.287 [-0.443, -0.121]
C-PR ($\lambda = 0$)	0.286 [0.166, 0.400]	0.222 [0.072, 0.366]	0.228 [0.063, 0.388]	0.058 [-0.108, 0.210]	0.029 [-0.117, 0.170]
C-PR ($\lambda = 0.25$)	0.284 [0.163, 0.399]	0.166 [0.011, 0.309]	0.168 [-0.002, 0.331]	0.008 [-0.157, 0.162]	-0.023 [-0.171, 0.117]
C-PR ($\lambda = 0.5$)	0.283 [0.162, 0.398]	0.129 [-0.018, 0.271]	0.131 [-0.041, 0.297]	-0.025 [-0.185, 0.129]	-0.062 [-0.210, 0.080]
C-PR ($\lambda = 0.75$)	0.284 [0.163, 0.398]	0.103 [-0.049, 0.254]	0.099 [-0.076, 0.267]	-0.051 [-0.212, 0.107]	-0.102 [-0.253, 0.047]
S-PR	0.304 [0.177, 0.418]	0.177 [0.033, 0.310]	0.146 [-0.031, 0.321]	0.009 [-0.143, 0.153]	-0.027 [-0.169, 0.111]

4.7.4 C-PR beside every other score

Rules A and B leave open how C-PR stands against the other scores of the study. Table 4.18 sets it against each of them and against the two raw degrees, on the spender labels of both windows and on the liquidation-free close rate of the W1 traders, with the resamples of the rules. Only the cells marked in the table belong to a rule; every other contrast was computed after the labels were known and decides nothing.

Table 4.18: Paired contrasts of C-PR ($\lambda = 0.5$) with every other score of this study and with the two raw degrees, on the spender labels of W0 ($n = 7{,}914$ contracts) and W1 ($n = 7{,}960$) and on the liquidation-free close rate of the W1 traders ($n = 70$). Each cell is $\Delta\tau = \tau_{\text{C-PR}} - \tau_b$ over its 95% interval; within a cohort every cell uses the same resamples. Cells marked [A] are the contrasts of rule A, fixed in the analysis plan; [B] marks those named in the registration of W1 (rule B with F5 and F6a) and [S] those of its first sensitivity analysis, with AWP as the comparator. These rules were fixed for the raw-unit analysis and are applied here to the revised scores. Every other cell was computed after the labels were known and belongs to no rule. The lower block repeats the contrasts with AWP and the in-approve degree on the spenders that had received a transfer from another address before the freeze, the spenders AWP can order; that restriction was also chosen after the results were known. CI = 95% percentile bootstrap over contracts (400 paired resamples, seed 42).

	W0 spenders		W1 spenders		W1 traders
C-PR minus	New approval pairs	New transfer senders	New approval pairs	New transfer senders	Liquidation-free close rate
C-PR ($\lambda = 1$)	+0.031[A] [0.021, 0.042]	+0.069 [0.057, 0.080]	+0.021[B] [0.009, 0.032]	+0.066 [0.054, 0.078]	-0.019[B] [-0.123, 0.089]
C-PR ($\lambda = 0$)	+0.151 [0.130, 0.170]	-0.030[A] [-0.040, -0.020]	+0.170[B] [0.149, 0.189]	-0.025[B] [-0.038, -0.015]	+0.000[B] [-0.010, 0.010]
EndorseRank	+0.026 [0.016, 0.037]	+0.064 [0.052, 0.075]	+0.017 [0.006, 0.029]	+0.064 [0.051, 0.076]	-0.015 [-0.116, 0.094]
EndorseRank (activity restarts)	-0.003 [-0.014, 0.008]	+0.066 [0.055, 0.078]	-0.019 [-0.029, -0.008]	+0.087 [0.075, 0.102]	-0.004 [-0.102, 0.105]
AWP	+0.133 [0.114, 0.153]	-0.073 [-0.083, -0.063]	+0.159[S] [0.139, 0.178]	-0.075[S] [-0.085, -0.065]	+0.019[S] [-0.026, 0.064]
S-PR	+0.139 [0.121, 0.156]	-0.041 [-0.049, -0.032]	+0.154 [0.134, 0.174]	-0.044 [-0.052, -0.037]	-0.017 [-0.094, 0.048]
In-approve degree at the freeze	-0.063 [-0.076, -0.052]	+0.052 [0.038, 0.065]	-0.065 [-0.077, -0.054]	+0.055 [0.040, 0.071]	-0.112 [-0.194, -0.026]
Transfer in-degree at the freeze	+0.126 [0.104, 0.145]	-0.111 [-0.121, -0.100]	+0.151 [0.130, 0.173]	-0.115 [-0.128, -0.105]	+0.030 [-0.021, 0.099]
Spenders that had received a transfer before the freeze (W0 $n = 6{,}489$, W1 $n = 6{,}567$)					
AWP	+0.008 [-0.002, 0.020]	-0.036 [-0.044, -0.026]	+0.022 [0.011, 0.033]	-0.037 [-0.046, -0.028]	—
In-approve degree at the freeze	-0.051 [-0.064, -0.037]	—	-0.053 [-0.066, -0.041]	—	—

Against EndorseRank, the walk over allowances alone, C-PR ranks both spender labels higher in

both windows: new approval pairs by 0.026 ($[0.016, 0.037]$) in W0 and 0.017 ($[0.006, 0.029]$) in W1, and new senders by 0.064 ($[0.052, 0.075]$) and 0.064 ($[0.051, 0.076]$). On the W1 traders it cannot be told apart from EndorseRank (-0.015, $[-0.116, 0.094]$). The variant of EndorseRank with activity restarts is level with C-PR on new approval pairs in W0 (-0.003, $[-0.014, 0.008]$) and ranks them higher in W1 (-0.019, $[-0.029, -0.008]$); on new senders C-PR leads it in both windows ($+0.066$, $[0.055, 0.078]$, and $+0.087$, $[0.075, 0.102]$). Within the window C-PR leads the variant on the transfer and Sybil-stability families (0.400 and 0.142 against 0.251 and 0.056) and trails it on the allowance family (0.449 against 0.601; Tables 4.3 and 4.8).

Against AWP, the walk over transfers alone, C-PR ranks new approval pairs higher ($+0.133$, $[0.114, 0.153]$, in W0 and $+0.159$, $[0.139, 0.178]$, in W1) and new senders lower (-0.073, $[-0.083, -0.063]$, and -0.075, $[-0.085, -0.065]$). Its lead on approvals comes largely from the spenders that AWP cannot order. AWP scores zero every spender without a transfer edge up to the freeze, 1,342 of them in W0, and those spenders gain new approval pairs more often than the rest (Section 4.6). Among the spenders that had received a transfer from another address before the freeze, C-PR and AWP are level on new approval pairs in W0 ($+0.008$, $[-0.002, 0.020]$) and C-PR leads in W1 ($+0.022$, $[0.011, 0.033]$), while AWP ranks new senders higher in both windows (-0.036, $[-0.044, -0.026]$, and -0.037, $[-0.046, -0.028]$) (lower block of Table 4.18; this restriction, too, was chosen after the results were known). Out of window, what C-PR adds to a walk over transfers is mainly an order among the contracts that this walk leaves tied, at the price of a worse order of later inflow among those it can rank.

Against the raw degrees, C-PR does not rank either label above the count of the same edge type. On new approval pairs it lies below the in-approve degree in both windows (-0.063, $[-0.076, -0.052]$, and -0.065, $[-0.077, -0.054]$); on new senders it is below the transfer in-degree in both windows (-0.111, $[-0.121, -0.100]$, and -0.115, $[-0.128, -0.105]$). Against its ablation S-PR it ranks new approval pairs higher ($+0.139$, $[0.121, 0.156]$, and $+0.154$, $[0.134, 0.174]$) and new senders lower (-0.041, $[-0.049, -0.032]$, and -0.044, $[-0.052, -0.037]$). On the liquidation-free close rate of the 70 traders C-PR ranks below the in-approve degree (-0.112, $[-0.194, -0.026]$), and no other

contrast of C-PR excludes zero.

Taken together, C-PR ranks every spender label above EndorseRank in both windows, and it ranks later approvals above AWP, largely through the contracts that AWP leaves tied. It stays below the raw count of each edge type on that edge type's label in both windows, below AWP on later senders, below EndorseRank with activity restarts on later approvals in W1, and below the in-approve degree on the traders' liquidation-free close rate. None of these contrasts was fixed in advance.

4.8 Outcomes built from neither edge type

The GMX trader labels are the only outcomes in these data built from neither edge type. The comparison of Section 3.7.3 was registered with rule B, before the study extracted the July–September logs and before it scored the liquidation labels of W0. Here its contrasts, rules and wording are applied to the revised scores as a re-analysis, with W1 as the primary window and W0 as the secondary one. Intervals come from 2,000 paired resamples and are 95% unless marked 97.5%, the Bonferroni interval of the decision contrasts P and Q. Table 4.19 gives the contrasts on the liquidation-free close rate, Table 4.20 the traders that receive approvals, and Table 8.11 (Appendix 8.5.4) all six trader labels.

R1 holds. In W1 the in-approve degree at the freeze reaches $\tau = 0.394$ ($[0.265, 0.518]$) on the liquidation-free close rate and the transfer in-degree 0.253 ($[0.117, 0.379]$), so P, their difference, is $+0.141$ (97.5% interval $[0.026, 0.256]$), an interval above zero. Stratified by the number of closes, P is $+0.145$ ($[0.040, 0.328]$). In W0, P is $+0.095$ (97.5% interval $[-0.041, 0.263]$), positive but not resolved; R1 asked for both W1 intervals above zero, so, in the registered wording, there is a count-level lead of allowance over transfer on later liquidation avoidance. R2 does not hold. Q, EndorseRank with AWP's activity restarts minus AWP (the like-for-like PageRank pair), is $+0.023$ in W1 and -0.012 in W0, and both 97.5% intervals include zero ($[-0.071, 0.127]$ and $[-0.177, 0.168]$), so the PageRank versions are not resolved. Neither are C-PR's layers ($+0.019$, $[-0.093, 0.128]$, in W1), EndorseRank against AWP ($+0.034$, $[-0.067, 0.134]$) or C-PR ($\lambda = 0$)

Table 4.19: Allowance-side against transfer-side scores on the liquidation-free close rate of the GMX V2 contract traders with at least three closes in the label window, a label built from neither edge type ($n = 70$ traders in W1, $n = 87$ in W0). The pairs, the decision contrasts P and Q with their Bonferroni 97.5% intervals and the stratified rows were registered before the study extracted the July–September logs, for the raw-unit analysis, and are applied here to the revised scores; W1 is the primary window and W0 the secondary one. Paired contract bootstrap with 2,000 resamples (seed 42). The stratified rows combine Kendall's τ_b within strata of the number of label-window closes (3–4, 5–9, 10–24, 25 or more), weighted by the number of contract pairs. Registered rules applied to the revised scores: R1 (count-level lead) met; R2 (PageRank-level lead) not met; R3 (specificity) applies.

Contrast (a minus b)	$\tau(a)$	$\tau(b)$	$\Delta\tau$	95% CI	97.5% CI
Window W1, July–September 2026 ($n = 70$ traders)					
In-approve degree minus transfer in-degree	0.394	0.253	0.141	[0.037, 0.240]	[0.026, 0.256]
EndorseRank (activity restarts) minus AWP	0.286	0.263	0.023	[-0.062, 0.113]	[-0.071, 0.127]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.302	0.282	0.019	[-0.093, 0.128]	–
EndorseRank minus AWP	0.297	0.263	0.034	[-0.067, 0.134]	–
C-PR ($\lambda = 0$) minus AWP	0.282	0.263	0.019	[-0.031, 0.066]	–
EndorseRank minus in-approve degree	0.297	0.394	-0.097	[-0.174, -0.009]	–
In-approve degree minus transfer in-degree, stratified by closes	0.381	0.236	0.145	[0.040, 0.328]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.273	0.238	0.034	[-0.096, 0.171]	–
Window W0, April–June 2026 ($n = 87$ traders)					
In-approve degree minus transfer in-degree	0.345	0.249	0.095	[-0.027, 0.234]	[-0.041, 0.263]
EndorseRank (activity restarts) minus AWP	0.242	0.254	-0.012	[-0.153, 0.149]	[-0.177, 0.168]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.237	0.294	-0.057	[-0.209, 0.092]	–
EndorseRank minus AWP	0.236	0.254	-0.019	[-0.173, 0.144]	–
C-PR ($\lambda = 0$) minus AWP	0.294	0.254	0.040	[-0.023, 0.098]	–
EndorseRank minus in-approve degree	0.236	0.345	-0.109	[-0.196, -0.018]	–
In-approve degree minus transfer in-degree, stratified by closes	0.342	0.253	0.089	[-0.025, 0.252]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.229	0.265	-0.036	[-0.186, 0.145]	–

against AWP ($+0.019$, $[-0.031, 0.066]$), in either window. In a contrast the registration did not name, EndorseRank falls below the approval count, by 0.097 ($[-0.174, -0.009]$) in W1 and 0.109 ($[-0.196, -0.018]$) in W0.

The other labels do not enter the decision (Table 8.11). On the no-liquidation flag, P is $+0.141$ ($[0.044, 0.232]$) in W1 and $+0.109$ ($[-0.022, 0.257]$) in W0. On profitable closes and realized gain, which grow with activity, the PageRank pairs favor the transfer side in W1, with intervals below zero for Q (-0.215, $[-0.399, -0.042]$, and -0.299, $[-0.457, -0.146]$) and for EndorseRank against AWP (-0.200, $[-0.390, -0.023]$, and -0.292, $[-0.460, -0.134]$); P points the same way without being resolved. R3 applies. In W1 the in-approve degree ranks below the transfer in-degree on the profitable share (-0.162, $[-0.295, -0.025]$) and the non-loss share (-0.222, $[-0.328, -0.106]$), so, in the registered wording, the signal concerns forced-liquidation avoidance, not trading success. Part of the non-loss gap is built in: the pipeline counts a liquidated close as a close that was not a loss, so a trader with liquidations gains non-loss share, and R3 therefore rests mainly on the profitable share, which counts a liquidated close as not profitable.

The count-level comparison rests on few traders. In W1 the highest in-approve degree among the traders is 1, so P in effect compares the 37 recipients with the 33 other traders; in W0 the maximum is 3, and 39 of the 87 traders are recipients. Every recipient avoided liquidation in both windows (37 of 37 and 39 of 39), against 72.7% of the other traders in W1 and 75.0% in W0. Recipients also closed more positions (median 24 against 11 in W1, 30 against 10 in W0) and had lower profitable and non-loss shares (mean non-loss share 0.433 against 0.644 in W1), the pattern behind R3. Without the 10 recipients of highest in-approve degree, P is $+0.155$ ($[0.032, 0.276]$) in W1 and $+0.088$ ($[-0.048, 0.240]$) in W0.

4.9 Claims (C1–C6)

The chapter's results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2. C1–C5 are the primary claims; C2 and C3 rest on contrasts com-

Table 4.20: GMX V2 contract traders with a positive in-approve degree at the freeze (recipients) against all other labelled traders, in both windows ($n = 70$ traders in W1, $n = 87$ in W0). The comparison of recipients with the other traders and the influence check were registered with the contrasts of Table 4.19 and are applied here to the revised scores. The highest in-approve degree among the traders is 1 in W1 and 3 in W0, so the ten recipients left out by the influence check are taken from tied values in node order. The influence check is a paired contract bootstrap on the remaining traders (2,000 resamples, seed 42).

	Recipients	Other traders
Window W1, July–September 2026 ($n = 70$ traders)		
Traders	37	33
Share with no liquidation	1.000	0.727
Mean liquidation-free close rate	1.000	0.878
Mean profitable share	0.325	0.479
Mean non-loss share	0.433	0.644
Median closes in the label window	24.0	11.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.155$ [0.032, 0.276]		
Window W0, April–June 2026 ($n = 87$ traders)		
Traders	39	48
Share with no liquidation	1.000	0.750
Mean liquidation-free close rate	1.000	0.919
Mean profitable share	0.359	0.474
Mean non-loss share	0.401	0.607
Median closes in the label window	30.0	10.0
P without the ten recipients of highest in-approve degree: $\Delta\tau = 0.088$ [-0.048, 0.240]		

puted after the labels were known, and C6 on the comparison on outcomes built from neither edge type. The verdicts of rules A and B in C1 and of R1–R3 in C6 apply the registered rules to the revised scores, as a re-analysis; the registered analysis reached the same verdicts except on R1 (Appendix 8.6).

C1. Under its rules, C-PR, a fixed-λ mixture of two dollar-weighted layers with uniform restarts, does not beat its two layers walked alone. Rule A fails: C-PR is no worse than the allowance walk on new approval pairs ($+0.031$, $[0.021, 0.042]$) but worse than the transfer walk on new senders (-0.030, $[-0.040, -0.020]$), and its same-window part fails as well (Tables 4.10 and 4.14). Rule B is not met (Table 4.16): the W0 gain over the transfer walk on new approval pairs, found after the labels were known, recurs in W1 (F1, $+0.170$, $[0.149, 0.189]$), but C-PR falls below the transfer walk on new senders by more than the margin of 0.01 (F2, -0.025, $[-0.038, -0.015]$), and F3 fails narrowly ($+0.000$, $[-0.010, 0.010]$). Against AWP, in the first sensitivity analysis, F2 fails clearly (-0.075, $[-0.085, -0.065]$). S-PR ranks new senders slightly above the transfer walk and new approval pairs well below the allowance walk, so allowance edges help mainly when they carry the walk; Section 4.7.1 reports where the hybrids stand within the window.

C2. Beside the rules, C-PR ranks both spender labels above EndorseRank in both windows (new approval pairs $+0.026$, $[0.016, 0.037]$, and $+0.017$, $[0.006, 0.029]$; new senders $+0.064$, $[0.052, 0.075]$, and $+0.064$, $[0.051, 0.076]$) and leads it within the window on the transfer and Sybil-stability families. It ranks new approval pairs above AWP ($+0.133$, $[0.114, 0.153]$; $+0.159$, $[0.139, 0.178]$), largely through the spenders that AWP scores zero: among the spenders that had received a transfer, the two are level in W0 and C-PR leads by 0.022 ($[0.011, 0.033]$) in W1. C-PR ranks new senders below AWP and below the transfer in-degree in both windows, new approval pairs below the in-approve degree in both windows (-0.063, $[-0.076, -0.052]$; -0.065, $[-0.077, -0.054]$) and below EndorseRank with activity restarts in W1 (-0.019, $[-0.029, -0.008]$), and the liquidation-free close rate of the W1

traders below the in-approve degree (-0.112, $[-0.194, -0.026]$) (Table 4.18). None of these contrasts was fixed in advance.

C3. Out of window neither single-layer walk beats the raw degree it smooths. AWP ranks later new transfer senders below the transfer in-degree in both windows (W0 -0.038, $[-0.043, -0.032]$; W1 -0.041, $[-0.047, -0.034]$), and EndorseRank ranks later new approval pairs below the in-approve degree in both (W0 -0.090, $[-0.100, -0.079]$; W1 -0.082, $[-0.090, -0.073]$) (Tables 4.10 and 4.13); in W0 neither comparison is a matter of coverage (Table 4.11), and no such restriction of these two contrasts was computed for W1. With activity restarts EndorseRank also stays below its count (Section 4.5.4), so on these labels propagation adds nothing to the count of the same edge type, whatever the restart rule. On the profit labels of the contract traders of W0 the paired contrasts separate the two sides only on realized gain, where the transfer layer walked alone ranks above the allowance layer (C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$) -0.257, $[-0.447, -0.071]$), not on profitable closes (EndorseRank minus AWP -0.046, $[-0.205, 0.113]$), and no score ranks the share of closes that were profitable (Tables 4.12 and 8.11). These comparisons were computed after the labels were known.

C4. In the same window each single-layer walk agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family (0.560, $[0.548, 0.571]$) and AWP with the transfer family (0.736, $[0.728, 0.744]$), and the mixtures of C-PR lie between their two layers (Table 4.3). These tests measure internal consistency, and part of the agreement is built in; external validity is outside their scope. The two walks order the cohort differently: their agreement, $\tau_b = 0.199$ ($[0.181, 0.221]$), has an interval that excludes zero and one; 96.4% of the contract pairs are ordered by both scores, and 39.8% of those are discordant. Neither result comes from ties or from the rule for contracts outside a graph (Table 4.4, Figure 4.3).

C5. On identical contract sets, the cost of each score's solver call grows with the size of its graph, since each iteration costs time in $|E|$ and $|V|$, and with the iterations its walk needs. The allowance graph has 45.2% of the transfer graph's edges, and AWP takes 3.21 times as long

as EndorseRank at an edge ratio of 2.21 (Table 4.1). On the subsamples the time ratio follows the edge ratio at no stage, and the stages do not test edge counts beyond those of the cohort graph (Table 4.2). The low cost is no advantage of the algorithm: C-PR and S-PR walk the transfer layer and take longer than AWP, C-PR 1.36 times as long.

C6. On later liquidation avoidance of the GMX V2 contract traders, the one outcome built from neither edge type that bears on counterparty risk, the approval count at the freeze leads the transfer count in the primary window W1 (P $= +0.141$, 97.5% interval $[0.026, 0.256]$), while in W0 P is positive but not resolved ($+0.095$, 97.5% interval $[-0.041, 0.263]$). The PageRank versions are not resolved (Q $= +0.023$, 97.5% interval $[-0.071, 0.127]$). In W1 the approval count ranks traders lower than the transfer count on the shares of closes that were profitable or not losing, so, in the wording of R3, the signal concerns forced-liquidation avoidance, not trading success; in W0 neither difference is resolved (Table 8.11). The samples are small, 70 and 87 contract traders, and every interval is wide (Section 4.8).

The same-window tables and the rank divergence answer RQ1, the benchmark tables RQ2, the holdouts of the single-layer walks with Table 4.18 RQ3, and Section 4.7 RQ4 and RQ5; C6 adds no research question. The direct comparisons between EndorseRank and AWP are in Appendix 8.5 and Table 4.19, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the research questions and objectives of Chapter 1.